

Education

- 2015-2020 **Dual Degree (B.Tech + M.Tech) in Computer Science and Engineering**
Indian Institute of Technology Madras, Chennai, India CGPA: 8.78
- 2015 **XII - Karnataka Board, KLE Society's Independent PU College, Bangalore** 97.30 %
- 2013 **X - ICSE, B P Indian Public School, Bangalore** 96.33%

Professional Experience

- Software Engineer, Level IV, Google LLC, New York**
- Dec 2022 - Present **Software Engineer, Level IV, Google India Pvt Ltd, Bangalore**
- Aug 2020 - Nov 2022 { Working on Keep, a notetaking editor in Google Workspace.
- { Developing intelligent features for the Google Workspace Editors (Docs, Slides, Keep, etc) using my expertise on the products' client-side software, supporting tools and libraries, and natural language processing infrastructure.
- { Using cutting-edge frontend tools like Web Assembly and Emscripten, and Google-internal technologies like j2Cl, client-side cross-platform frameworks and build systems, to develop user-facing features such as spellcheck in encrypted documents for five languages and writing style suggestions for English text.
- { Formulating technical designs for independent end-to-end problems, driving cross-team collaboration, upholding software reliability practices, technical-debt resolution and documentation, and proactively identifying areas of future work.
- { Guiding junior engineers on programming and software design tasks to enable timely delivery of products to customers.
- May 2019 - July 2019 **Software Engineering Intern, Google India Pvt Ltd, Bangalore**
- Worked on the Editors client-side software infrastructure to develop a user interface with control options to undo or provide feedback on the correction and a logging framework, for the Google Docs text auto-correction feature.
- May 2018 - July 2018 **Research Intern, Big Data Experience Labs, Adobe Research, Bangalore**
- Developed a mobile application for Text to Scene Conversion in Augmented Reality, based on novel research techniques for prediction of three-dimensional object sizes and positions from textual features.

Research Experience

- Sep 2019 - May 2020 **Paraphrase Generation with a Bilingual Model and Continuous Embeddings** *Master's Thesis, Language Technologies Institute, Carnegie Mellon University*
- Machinated a novel technique for paraphrase generation using the [von Mises-Fisher \(vMF\) Loss](#) on a transformer network, and showed that it produces superior paraphrases as compared to the log-likelihood model by employing bilingual data to induce zero-shot paraphrasing, guided by [Prof. Yulia Tsvetkov](#).
- May 2017 - July 2017 **Cognitive Approach to Natural Language Processing**
- Research Intern, Department of Computer Science and Automation, Indian Institute of Science (IISc), Bangalore*
- Developed a cognitive text parser that combines syntactic and semantic approaches, to process textual data into cognitive structural representations, to be used as a feature extractor for downstream NLP tasks, and demonstrated the correlation of the extracted cognitive features with semantic and syntactic text features, guided by [Prof. Veni Madhavan](#).

Publications and Patents

- [Publication and Poster] **Improving the Diversity of Unsupervised Paraphrasing with Embedding Outputs (Paper, Poster)**
Monisha Jegadeesan, Sachin Kumar, John Wieting, Yulia Tsvetkov
In [Workshop on Multilingual Representation Learning](#),
The 2021 Conference on Empirical Methods in Natural Language Processing ([EMNLP 2021](#))
- [Publication and Poster] **Adversarial Demotion of Gender Bias in Natural Language Generation (Paper, Poster)**
Monisha Jegadeesan
In [ACM CODS-COMAD 2020](#) - Young Researchers' Symposium
- [Poster] **ARComposer: Authoring Augmented Reality Experiences through Text (Poster)**
Sumit Kumar, Paridhi Maheshwari, Monisha Jegadeesan, Amrit Singhal, Kush Kumar Singh, Kundan Krishna
In ACM User Interface Software and Technology Symposium 2019 ([ACM UIST 2019](#))
- [Filed Patent] **Visualizing Natural Language through 3D Scenes in Augmented Reality**
Sumit Kumar, Paridhi Maheshwari, Monisha Jegadeesan, Amrit Singhal, Kush Kumar Singh, Kundan Krishna
Filed at the US PTO (Application Number: 16/247,235)

Projects

Real-Time AI Chatbot for Customer Support (2024)

Developed and validated large-scale training datasets to build an AI-powered chatbot capable of answering customer queries by leveraging company knowledge bases. Structured JSON tasks were created to implement function-calling workflows using LangChain, ensuring data consistency and workflow automation aligned with agentic AI system requirements. Collaborated closely with AI researchers to enhance dataset quality and accuracy, resulting in a 35% reduction in human agent workload and instant query responses, demonstrating strong data validation and JSON expertise.

Tech: Python, LangChain, OpenAI API, ChromaDB, JSON

Customer Sentiment Analysis with NLP (2024)

Validated and manipulated raw textual datasets from e-commerce product reviews to classify sentiment using Logistic Regression. Utilized Python scripts and SQL queries to clean, structure, and query datasets, ensuring data quality and consistency for effective sentiment modeling. Created structured JSON output to facilitate function-calling tasks for downstream agentic AI workflows. This led to an 89% sentiment classification accuracy and identification of critical product improvement areas, highlighting skills in data handling, annotation, and NLP best practices.

Tech: Python, NLTK, SQL, JSON, Logistic Regression

Call Center Performance Analytics Dashboard (2024)

Built and maintained high-quality, structured datasets from call center logs by writing optimized SQL queries to clean, aggregate, and validate data focused on KPIs such as agent performance, wait time, and SLA compliance. Developed interactive dashboards with Power BI using DAX to monitor data accuracy and integrity, enabling data-driven improvements including a 22% reduction in wait times and enhanced SLA compliance. Produced documentation of data workflows and best practices to support consistency across projects aligned with AI operational standards.

Tech: SQL, Power BI, Excel, DAX, JSON

Predictive Maintenance Model for Industrial IoT (2024)

Engineered and validated large-scale sensor datasets comprising temperature, vibration, and pressure features for predictive maintenance using Python and Random Forest algorithms. Performed feature extraction and dataset quality assurance to ensure consistency and alignment with frontier AI requirements for time-series modeling. Generated structured JSON tasks to support function-calling mechanisms in AI workflows, contributing to a 28% reduction in unexpected machine downtime and significant cost savings. Emphasized data quality, workflow automation, and AI collaboration throughout the project.

Tech: Python, Random Forest, Sensor Data Processing, JSON

About :

I specialize in validating and ensuring the quality of large-scale training datasets for agentic AI systems, leveraging strong proficiency in Python and JSON handling to create and manipulate structured tasks for function-calling workflows. Experienced in maintaining data accuracy, consistency, and integrity, and skilled in SQL for querying and managing datasets to support cutting-edge AI requirements. Passionate about collaborating with AI researchers to align datasets with advanced LLMs, NLP best practices, and agentic AI use cases. Committed to developing robust data workflows and contributing to documentation and best practices in dataset preparation and validation.

